

Ludus and GOAD Pentest Environment

 th4ntis.com/guide/2025/09/09/Ludus-and-GOAD-Pentest-Environment.html

Th4ntis:~\$

September 9, 2025

This is my guide on setting up [Game Of Active Directory\(GOAD\)](#) inside of [Ludus](#).

About

This home pentesting environment can come at a small price point, smaller than I originally thought I would need. You can find mini-pcs pretty cheap on eBay or on Facebook Marketplace in some cases. Just upgrade them a bit with more memory and potentially more storage, and you're set.

[Ludus](#)

[Ludus](#) is a system to build easy to use cyber environments, or "ranges" for testing and development. Built on Proxmox, Ludus enables advanced automation while still allowing easy manual modifications or setup of virtual machines and networks. Ludus is implemented as a server that runs Packer and Ansible to create templates and deploy complex cyber environments from a single configuration file. Ludus is accessed via the Ludus CLI (client) or the Proxmox web interface. Normal users should not need to access Ludus via SSH.

There are also other [environments guides](#), such as setting up an [ADCS](#), [Malware Lab](#), [SCCM](#), and more to deploy on Ludus.

[GOAD](#)

[Game Of Active Directory](#) is a free pentest active directory LAB(s) project. The purpose of this tool is to give pentesters a vulnerable Active directory environment ready to use to practice usual attack techniques. The idea behind this project is to give you an environment where you can try and train your pentest skills without having the pain to build all by yourself.

Ludus

Requirements

Ludus does have specific system requirements to be used. I've highlighted some specifics.

Getting Started

Ludus is *only* compatible with virtualization capable Debian 12 or Proxmox 8. No other environments or hosting solutions will be supported.

Requirements

- x86_64 (aka amd64 aka 64-bit "Intel") CPU with a Passmark score > 6,000
- A host capable of virtualization running Debian 12 or Proxmox 8 (e.g. "bare metal", Azure Dv3 and Ev3, AWS *.metal, supported Google Cloud VMs see Deployment Options)
- at least 32GB RAM per user/range that will be deployed
- at least 200GB storage for initial templates and at least 50GB per user/range that will be deployed (large, fast NVMe drives recommended)
- no more than 150 users per Ludus host
- Internet connected (not via WiFi)
- Root shell access
- If you want to access Ludus across the internet
 - 1 public IP address
 - the ability to allow in arbitrary ports (i.e. port forwarding or control of the cloud firewall)

Machines with lower specs than listed above may work, but are not tested/supported.

Starting out

I am running this on a Lenovo ThinkCentre M70Q with:

- [Debian 12](#)
- 64GB RAM
- A 1TB NVMe drive
- Intel-i5-10400T

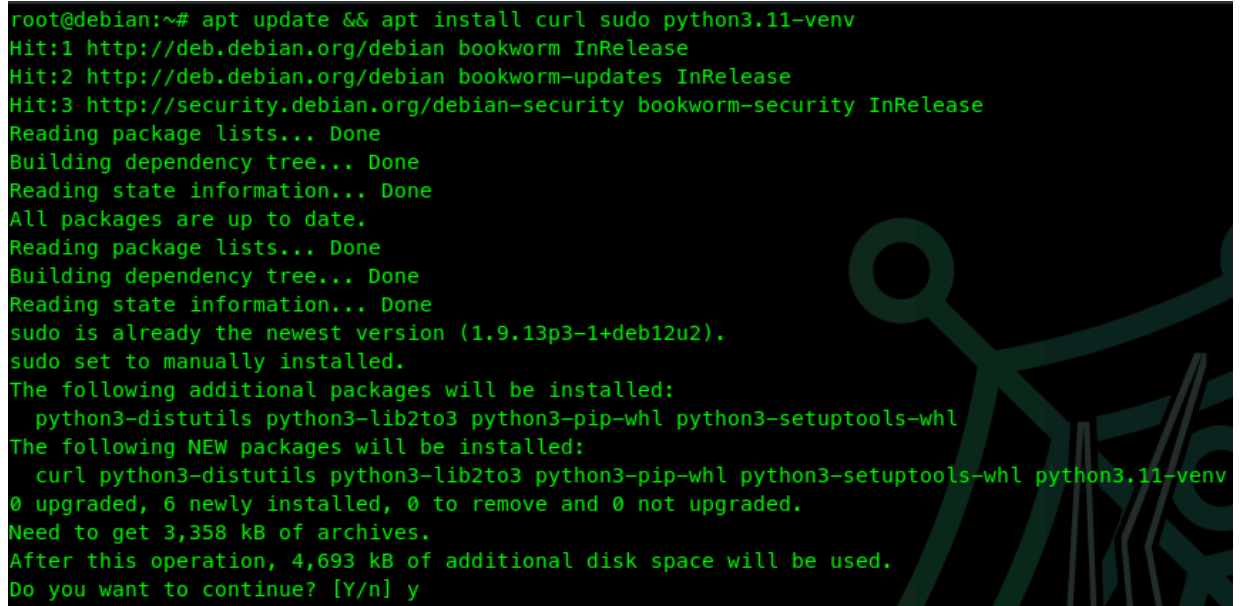
Installing

1. After installing [Debian 12](#), while on the machine physically or accessing remotely, change to the root user

```
su -
```

2. Install the reqs

```
apt update && apt install curl sudo python3.11-venv
```



```
root@debian:~# apt update && apt install curl sudo python3.11-venv
Hit:1 http://deb.debian.org/debian bookworm InRelease
Hit:2 http://deb.debian.org/debian bookworm-updates InRelease
Hit:3 http://security.debian.org/debian-security bookworm-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
sudo is already the newest version (1.9.13p3-1+deb12u2).
sudo set to manually installed.
The following additional packages will be installed:
  python3-distutils python3-lib2to3 python3-pip-whl python3-setuptools-whl
The following NEW packages will be installed:
  curl python3-distutils python3-lib2to3 python3-pip-whl python3-setuptools-whl python3.11-venv
0 upgraded, 6 newly installed, 0 to remove and 0 not upgraded.
Need to get 3,358 kB of archives.
After this operation, 4,693 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
```

3. Install Ludus - This will go through and ask a name for the node within Proxmox, the network interface you want to use, etc. As I set a static IP for my machine already, I am going to keep the information already populated and just enter through. This will restart the machine twice. I would give this about 15-20 min.

```
curl -s https://ludus.cloud/install | bash
```

```
root@debian:~# curl -s https://ludus.cloud/install | bash
=====
[+] Client install prefix set to /usr/local/bin
[+] Created temp dir at /tmp/ludus-client.DViwe8
[+] Architecture detected as x86_64
[+] OS detected as Linux
[+] Downloaded ludus-client_linux-amd64-1.10.0 into /tmp/ludus-client.DViwe8
[+] Downloaded ludus checksums file into /tmp/ludus-client.DViwe8
[+] Checksum of /tmp/ludus-client.DViwe8/ludus-client_linux-amd64-1.10.0 verified
[+] Install prefix already exists. No need to create it.
[+] Installed ludus-client_linux-amd64-1.10.0 to /usr/local/bin/ as 'ludus'
[+] Ludus client installation complete
[+] Shell completions already installed
[?] Would you like to install the Ludus server on this host?
[?] (y/n): y
```

```
root@debian:~# curl -s https://ludus.cloud/install | bash
=====
[+] Client install prefix set to /usr/local/bin
[+] Created temp dir at /tmp/ludus-client.DViwe8
[+] Architecture detected as x86_64
[+] OS detected as Linux
[+] Downloaded ludus-client_linux-amd64-1.10.0 into /tmp/ludus-client.DViwe8
[+] Downloaded ludus checksums file into /tmp/ludus-client.DViwe8
[+] Checksum of /tmp/ludus-client.DViwe8/ludus-client_linux-amd64-1.10.0 verified
[+] Install prefix already exists. No need to create it.
[+] Installed ludus-client_linux-amd64-1.10.0 to /usr/local/bin/ as 'ludus'
[+] Ludus client installation complete
[+] Shell completions already installed
[?] Would you like to install the Ludus server on this host?
[?] (y/n): y
[+] Installing Ludus server
[+] Downloaded ludus-server-1.10.0 into /tmp/ludus-client.DViwe8
[+] Checksum of /tmp/ludus-client.DViwe8/ludus-server-1.10.0 verified
ludus server 1.10.0+70c2021
Extracting ludus to /opt/ludus...
ludus files extracted successfully

Only run Ludus install on a clean Debian 12
machine that will be dedicated to Ludus

[+] Understand [x] Exit
```

=====

LOGOS

Ludus Interactive Installer //////////////////////////////////////

```
What is the public IP of this host
This is used as the WireGuard endpoint
> 192.168.1.30
```

=====

LOGOS

```
[+] Client install prefix set to /usr
```

```
[+] Created temp dir at /tmp/ludus-c
```

```
[*] created temp dir at /tmp/cadd3-c
```

[+] Architecture detecte
[-] Architecture not detecte

```
[+] OS detected as Linux
```

```
[+] Downloaded ludus-client_linux-amd64-1.10.0 into /tmp/ludus-cl
```

```
[+] Downloaded ludus checksums file into /tmp/ludus-client.DViwe8
```

```
[+] Checksum of /tmp/ludus-client.DViwe8/ludus-client_li
```

```
[+] Install prefix already exists. No need to create it.
```

```
[+] Installed ludus-client linux-amd64
```

```
[+] ludus client installation complete
```

```
[*] Euduo client installation complete
[*] Shell completions already installed
```

[4] SHELL CO.

[?] Would you like to insta

```
[?] (y/n): y
```

[+] Installing Ludus server

[+] Downloaded ludus-server-

```
[+] Checksum of /tmp/ludus-client.DViwe8/ludus-server-1.10.0 ver
```

Ludus server 1.10.0+70c2d21

Extracting ludus to /opt/ludus

```

Extracting 'caddis' to 'opt/caddis...
Ludus files extracted successfully

```

Ludus Interactive Installer //////////////////////////////////////

The suggestion is usually correct

```
> 192.168.1.1
```

The suggestion is usually correct

$> 255,255,255.0$

```
> local
```

```
> qcow2
```

```
proxmox_node: ludus
proxmox_interface: eno1
proxmox_local_ip: 192.168.1.30
proxmox_public_ip: 192.168.1.30
```

=====

LEADS

```
Ludus Interactive Installer //////////////////////////////////////
```

```
> local
```

Allow

```
proxmox_node: ludus
proxmox_interface: eno1
proxmox_local_ip: 192.168.1.30
proxmox_public_ip: 192.168.1.30
proxmox_gateway: 192.168.1.1
proxmox_netmask: 255.255.255.0
proxmox_vm_storage_pool: local
proxmox_vm_storage_format: qcow2
```

=====

L O O K S

- **Ludus install** will cause the machine to reboot twice.
- Install will continue automatically after each reboot.
- Check the progress of the install by running:
'ludus-install-status' from a root shell.

```
Installing ansible with pip
  Updating apt cache...
  Installing python3-pip...
  Installing ansible with pip...
```

30%

```

This is a Ludus host.

Ludus is a project to enable teams to quickly and
safely deploy test environments (ranges) to test tools and
techniques against representative virtual machines.

Ludus leverages Proxmox which is licensed under APGLv3.

Docs:   https://192.168.1.30:8080/ludus
Web UI: https://192.168.1.30:8006

Ludus Version: 1.10.0+70c2d21

```

Making an Admin User

1. Once installed, find the API Key. **** WRITE THIS DOWN ****

```
ludus-install-status
```

```

root@ludus:~# ludus-install-status
Ludus install completed successfully
Root API key: [REDACTED]
root@ludus:~#

```

2. Make new admin user and get the new admin users API key. **WRITE THIS DOWN.**
Replace 'API-KEY' with the root API key from the previous command.

```

LUDUS_API_KEY='API-KEY' \
ludus user add --name "USER" --userid USER --admin --url
https://127.0.0.1:8081

```

3. Set the API key for the newly created admin user

```
export LUDUS_API_KEY='API-KEY-OF-NEW-ADMIN-USER'
```

4. Getting Proxmox password for new user. **** WRITE THIS DOWN ****

```
ludus user creds get
```

```

root@ludus:~# LUDUS_API_KEY='[REDACTED]' ludus user add --name "Th4ntis Admin" --userid Th4ntisAdmin --admin --url https://127.0.0.1:8081
[INFO] Adding user to Ludus, this can take up to a minute. Please wait.

```

USERID	PROXMOX USERNAME	ADMIN	API KEY
Th4ntisAdmin	th4ntis-admin	true	[REDACTED]

```

root@ludus:~# export LUDUS_API_KEY='[REDACTED]'
root@ludus:~# ludus user creds get

```

PROXMOX USERNAME	PROXMOX PASSWORD
th4ntis-admin	[REDACTED]

Building Templates for GOAD environment

1. View current templates

```
ludus templates list
```

```
root@ludus:~# ludus templates list
+-----+-----+
|          TEMPLATE          | BUILT |
+-----+-----+
| debian-11-x64-server-template | FALSE |
| debian-12-x64-server-template | FALSE |
| kali-x64-desktop-template     | FALSE |
| win11-22h2-x64-enterprise-template | FALSE |
| win2022-server-x64-template   | FALSE |
+-----+-----+
root@ludus:~#
```

2. Get new templates needed for GOAD.

```
git clone https://gitlab.com/badsectorlabs/ludus && cd ludus/templates
ludus templates add -d win2019-server-x64
ludus templates add -d win2016-server-x64
```

```
root@ludus:~/ludus/templates# ludus templates add -d win2019-server-x64
[INFO] Successfully added template
root@ludus:~/ludus/templates# ludus templates add -d win2016-server-x64
[INFO] Successfully added template
root@ludus:~/ludus/templates# ludus templates list
+-----+-----+
|          TEMPLATE          | BUILT |
+-----+-----+
| debian-11-x64-server-template | FALSE |
| debian-12-x64-server-template | FALSE |
| kali-x64-desktop-template     | FALSE |
| win11-22h2-x64-enterprise-template | FALSE |
| win2022-server-x64-template   | FALSE |
| win2016-server-x64-template   | FALSE |
| win2019-server-x64-template   | FALSE |
+-----+-----+
root@ludus:~/ludus/templates# cd
root@ludus:~#
```

3. Build the templates - This will take a while

```
ludus templates build
```

4. While it builds, you can check status of builds with:

```
ludus templates logs -f
```

```
root@ludus:~# ludus templates build
[INFO] Template building started - this will take a while. Building 1 template(s) at a time.
root@ludus:~# ludus templates logs -f
2025/08/03 21:33:55 ui: ==> proxmox-iso.debian11: Retrieving ISO
2025/08/03 21:33:55 ui: ==> proxmox-iso.debian11: Trying https://cdimage.debian.org/cdimage/archive/11.7.0/amd64/iso-cd/debian-11.7.0-amd64-netinst.iso
2025/08/03 21:33:55 ui: ==> proxmox-iso.debian11: Trying https://cdimage.debian.org/cdimage/archive/11.7.0/amd64/iso-cd/debian-11.7.0-amd64-netinst.iso?checksum=sha512%3A4480ef6470f6d8ae193c268e213d33a6a5a8da90c20
38c1824784faa4e4473f0c9b546a4e2d34c43fbbd43542ae4fb93cfd5cb8ac9b88a476f1a6877c478674
```

Once the templates are installed, you will see it say it's complete.

```
==> Wait completed after 28 minutes 7 seconds
==> Builds finished. The artifacts of successful builds are:
=>=====
=> Build complete!
=>=====
^C
root@ludus:~# ludus templates list
+-----+-----+
|          TEMPLATE          | BUILT |
+-----+-----+
| debian-11-x64-server-template | TRUE  |
| debian-12-x64-server-template | TRUE  |
| kali-x64-desktop-template     | TRUE  |
| win11-22h2-x64-enterprise-template | TRUE  |
| win2022-server-x64-template   | TRUE  |
| win2016-server-x64-template   | TRUE  |
| win2019-server-x64-template   | TRUE  |
+-----+-----+
root@ludus:~#
```

GOAD

1. Clone the GOAD repository

```
git clone https://github.com/Orange-Cyberdefense/GOAD.git && cd GOAD
```

2. Start GOAD, check that everything is set with no errors, and install. This will take a long time.

```
./goad.sh -p ludus
```

3. Exit GOAD and update /root/.goad/goad.ini with your users API key

```
GOAD/ludus/local > check
```

```
GOAD/ludus/local > set_lab XXX # GOAD/GOAD-Light/NHA/SCCM
```

```
GOAD/ludus/local > install
```

```

root@ludus:~# git clone https://github.com/Orange-Cyberdefense/GOAD.git && cd GOAD
Cloning into 'GOAD'...
remote: Enumerating objects: 7258, done.
remote: Counting objects: 100% (945/945), done.
remote: Compressing objects: 100% (290/290), done.
remote: Total 7258 (delta 750), reused 655 (delta 655), pack-reused 6313 (from 3)
Receiving objects: 100% (7258/7258), 24.89 MiB | 41.11 MiB/s, done.
Resolving deltas: 100% (3478/3478), done.
root@ludus:~/GOAD# ./goad.sh -p ludus
Python version in use : 3.11.2
python version >= 3.8 ok
venv module is installed. continue
[+] venv not found, start python venv creation

```

```

GOAD
Game Of Active Directory
Pwning is coming

goad management console type help or ? to list commands

[*] goad config file not found, create file /root/.goad/goad.ini
[*] Start Loading default instance
[*] Lab instances :
[-] No instance found, change your config and use install to create a lab instance

GOAD/ludus/local > check
[*] ludus found in PATH
[*] Using api key from env
[*] CWD:
[*] Running command : ludus version
[*] Api key is set
[*] Using api key from env
[*] CWD:
[*] Running command : ludus --url https://127.0.0.1:8881 user list --json
[{"name":"th4ntis ludus","userID":"TL","dateCreated":"2025-08-03T21:29:45:276379448-04:00","dateLastActive":"2025-08-04T08:49:43:729148808-04:00","isAdmin":true,"proxmoxUsername":"th4ntis-ludus","portforwardingEnabled":false}]
[*] Current user name : th4ntis ludus
[*] Current user ID : TL
[*] User is admin : True
[*] ansible-playbook found in PATH
[*] Ansible galaxy collection ansible.windows is installed
[*] Ansible galaxy collection community.general is installed
[*] Ansible galaxy collection community.windows is installed

GOAD/ludus/local >

```

```

GOAD/ludus/local > install
[*] Current Settings
[*] Current Lab : GOAD
[*] Current Provider : ludus
[*] Current Provisioner : local
[*] Extension(s) :

Create lab with these settings ? (y/N)y
[*] Create instance folder
[*] Create instance providing files
[*] Instance vagrantfile created : /workspace/7936c8-goad-ludus/provider/config.yml
[*] Create lab provisioning file inventory_disable_vagrant
[*] Lab inventory file created : /workspace/7936c8-goad-ludus/inventory_disable_vagrant
[*] Create instance provisioning files
[*] Instance inventory file created : /workspace/7936c8-goad-ludus/inventory
[*] Create instance extensions inventory files
[*] Instance 7936c8-goad-ludus created
[*] Instance 7936c8-goad-ludus loaded



| Instance ID         | Lab  | Provider | IP Range        | Status       | Is Default | Extensions |
|---------------------|------|----------|-----------------|--------------|------------|------------|
| > 7936c8-goad-ludus | GOAD | ludus    | 192.168.56.0/24 | not provided | No         |            |



[*] Launch providing
[*] Using api key from env
[*] CWD: /workspace/7936c8-goad-ludus/provider
[*] Running command : ludus version
[*] Api key is set
[*] Using api key from env
[*] CWD: /workspace/7936c8-goad-ludus/provider
[*] Running command : ludus --url https://127.0.0.1:8881 user list --json
[{"name":"th4ntis ludus","userID":"TL","dateCreated":"2025-08-03T21:29:45:276379448-04:00","dateLastActive":"2025-08-04T08:58:33:715267538-04:00","isAdmin":true,"proxmoxUsername":"th4ntis-ludus","portforwardingEnabled":false}]
[*] Current user name : th4ntis ludus
[*] Current user ID : TL
[*] User is admin : True
[*] Using api key from env
[*] CWD: /workspace/7936c8-goad-ludus/provider
[*] Running command : ludus --url https://127.0.0.1:8881 user list all --json
[*] Lab user does not exist create it
[*] Using api key from env
[*] CWD: /workspace/7936c8-goad-ludus/provider
[*] Running command : ludus --url https://127.0.0.1:8881 user add -n [redacted] -l [redacted]

```

I did have a couple error that stopped the install, so added the API of my admin user key to /root/.goad/goad.ini then started the install again.

```

./goad.sh -p ludus
GOAD/ludus/local > install

```

```
GOAD/ludus/local > ls
```

Instance ID	Lab	Provider	IP Range	Status	Is Default	Extensions
> 7b0df7-goad-ludus	GOAD	ludus	192.168.56.0/24	not provided	Yes	

```
GOAD/ludus/local > load 7b0df7-goad-ludus
```

```
[+] Instance 7b0df7-goad-ludus loaded
```

Instance ID	Lab	Provider	IP Range	Status	Is Default	Extensions
> 7b0df7-goad-ludus	GOAD	ludus	192.168.56.0/24	not provided	Yes	

```
GOAD/ludus/local (7b0df7-goad-ludus) > install
```

```
PLAY RECAP *****
```

```
dc01      : ok=15   changed=4   unreachable=0   failed=0   skipped=1   rescued=0   ignored=0
dc02      : ok=31   changed=14  unreachable=0   failed=0   skipped=6   rescued=0   ignored=0
dc03      : ok=16   changed=7   unreachable=0   failed=0   skipped=0   rescued=0   ignored=0
srv02     : ok=18   changed=10  unreachable=0   failed=0   skipped=1   rescued=0   ignored=0
srv03     : ok=17   changed=11  unreachable=0   failed=0   skipped=1   rescued=0   ignored=0
```

```
[*] Lab successfully provisioned in 20:10:25
```

```
GOAD/ludus/local (7b0df7-goad-ludus) > ls
```

Instance ID	Lab	Provider	IP Range	Status	Is Default	Extensions
> 7b0df7-goad-ludus	GOAD	ludus	10.3.10.0/24	installed	Yes	

Then once more when it was using the API key from the environment and deploying, so I had to load the instance from within GOAD.

```
./goad.sh
```

```
GOAD/ludus/local > ls
```

```
GOAD/ludus/local > load (INSTANCE-ID)
```

```
GOAD/ludus/local > install
```

```
[!] Error during deployment
```

```
[*] Using api key from env
```

```
[*] CWD: /workspace/a7dfb9-goad-ludus/provider
```

```
[*] Running command : ludus --user GOADa7dfb9 range errors
```

```
***** ERROR 1 *****
```

```
fatal: [GOADa7dfb9-GOAD-DC03]: UNREACHABLE! => {"changed": false, "msg": "ssl: HTTPSConnectionPool(host='goada7dfb9-goad-dc03', port=5986): Max retries exceeded with url: /wsman (Caused by NewConnectionError('<urllib3.connection.HTTPSConnection object at 0x7cbff0f57c50>: Failed to establish a new connection: [Errno -2] Name or service not known'))", "unreachable": true}
```

```
***** ERROR 2 *****
```

```
fatal: [GOADa7dfb9-GOAD-SRV02]: UNREACHABLE! => {"changed": false, "msg": "winrm connection error: HTTPSConnectionPool(host='10.3.10.22', port=5986): Read timed out. (read timeout=320)", "unreachable": true}
```

```
***** ERROR 3 *****
```

```
fatal: [GOADa7dfb9-GOAD-SRV03]: UNREACHABLE! => {"changed": false, "msg": "ssl: HTTPSConnectionPool(host='goada7dfb9-goad-srv03', port=5986): Max retries exceeded with url: /wsman (Caused by NewConnectionError('<urllib3.connection.HTTPSConnection object at 0x7cbff0de2c50>: Failed to establish a new connection: [Errno -2] Name or service not known'))", "unreachable": true}
```

```
[!] Providing error stop
```

```
GOAD/ludus/local (a7dfb9-goad-ludus) > install
```

Kali VM with Ludus and GOAD

1. Find the GOAD UserID

```
ludus users list all
```

```
root@ludus:~/GOAD# ludus users list all
```

NAME	USERID	CREATED	LAST ACTIVE	ADMIN
root	ROOT	2025-08-03 21:23	2025-08-03 21:29	true
th4ntis ludus	TL	2025-08-03 21:29	2025-08-04 11:15	true
GOAD7b0df7	GOAD7b0df7	2025-08-04 08:58	2025-08-04 08:58	false

2. Get the config.yml file

```
ludus --user GOAD-USERID range config get > basic-config.yml
```

```
root@ludus:~# ludus --user GOAD7b0df7 range config get > config.yml
root@ludus:~# ls
config.yml GOAD ludus
root@ludus:~#
```

3. Edit the basic-config.yml file with the following added to the bottom

```
- vm_name: "-kali"
hostname: "-kali"
template: kali-x64-desktop-template
vlan: 10
ip_last_octet: 99
ram_gb: 4
cpus: 4
linux: true
testing:
  snapshot: false
  block_internet: false
```

4. Set the new config file and deploy it

```
ludus --user GOAD-USERID range config set -f basic-config.yml
ludus --user GOAD-USERID range deploy -t vm-deploy
```

5. Check on the deployment process

```
ludus --user GOAD-USERID range logs -f
```

```
root@ludus:~# ludus --user G0AD7b0df7 range config set -f config.yml
[INFO] Your range config has been successfully updated.
root@ludus:~# ludus --user G0AD7b0df7 range deploy -t vm-deploy
[INFO] Range deploy started
root@ludus:~# ludus --user G0AD7b0df7 range logs -f

PLAY [Pre run checks] *****

TASK [Acquire session ticket] *****
ok: [localhost]

TASK [Extract ticket from response] *****
ok: [localhost]

TASK [Check for valid dynamic inventory] *****
ok: [localhost] => {
  "changed": false,
  "msg": "Dynamic inventory loaded!"
}

PLAY RECAP *****
localhost : ok=52  changed=6  unreachable=0  failed=0  skipped=56  rescued=0  ignored=0

^C
```

6. Deploy the Kali VM

```
ludus --user GOAD-USERID range deploy --limit localhost,G0AD-USERID-kali
```

```
root@ludus:~# ludus --user G0AD7b0df7 range deploy --limit localhost,G0AD7b0df7-kali
[INFO] Range deploy started
root@ludus:~# ludus --user G0AD7b0df7 range logs -f

PLAY [Pre run checks] *****

TASK [Acquire session ticket] *****
ok: [localhost]

TASK [Extract ticket from response] *****
ok: [localhost]

TASK [Check for valid dynamic inventory] *****
ok: [localhost] => {
  "changed": false,
  "msg": "Dynamic inventory loaded!"
}

PLAY RECAP *****
G0AD7b0df7-kali : ok=20  changed=9  unreachable=0  failed=0  skipped=20  rescued=0  ignored=0
localhost      : ok=63  changed=3  unreachable=0  failed=0  skipped=98  rescued=0  ignored=0
```

7. Verify all VMs with their status

```
ludus --user GOAD-USERID range status
```

```
root@ludus:~# ludus --user GOAD6f0f0b range status
```

USER ID	RANGE NETWORK	LAST DEPLOYMENT	NUMBER OF VMS	DEPLOYMENT STATUS	TESTING ENABLED
GOAD6f0f0b	10.3.0.0/16	2025-08-07 08:13	7	Success	FALSE

PROXMOX ID	VM NAME	POWER	IP
107	GOAD6f0f0b-router-debian11-x64	On	10.3.10.254
108	GOAD6f0f0b-GOAD-DC01	On	10.3.10.10
109	GOAD6f0f0b-GOAD-DC02	On	10.3.10.11
110	GOAD6f0f0b-GOAD-DC03	On	10.3.10.12
111	GOAD6f0f0b-GOAD-SRV02	On	10.3.10.22
112	GOAD6f0f0b-GOAD-SRV03	On	10.3.10.23
113	GOAD6f0f0b-kali	On	10.3.10.99

8. Get Wireguard configuration file

```
ludus user wireguard | tee ludus.conf
```

```
root@ludus:~# ludus user wireguard | tee ludus.conf
[Interface]
# ...
[Peer]
# ...
root@ludus:~#
```

9. Save the file as a .conf file and use Wireguard to connect to it. Once connected, access your Kali machine via GUI in the browser, over SSH.

```
sudo wg-quick up GOAD.conf
```



```
login as: kali
kali@10.3.10.99's password:
Linux GOAD6f0f0b-kali 6.12.33+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.12.33-lka
lil (2025-06-25) x86_64

The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Thu Aug  7 08:13:55 2025 from 192.0.2.254
-- (Message from Kali developers)
|
| This is a minimal installation of Kali Linux, you likely
| want to install supplementary tools. Learn how:
| = https://www.kali.org/docs/troubleshooting/common-minimum-setup/
|
-- (Run: "touch ~/.hushlogin" to hide this message)
(kali@GOAD6f0f0b-kali)~$
```

I will be posting how to attack this environment soon! Keep an eye out!
